Eisenstein congruence in Iwasawa theory

(Iwasawa theory for unitary groups: the method of Eisenstein congruences and applications)

Ruichen XU

University of Vienna

October 13, 2026

I. Iwasawa theory of class groups – a gentle introducion

I. Iwasawa theory of class groups: number fields and class groups

Let F be a number field, that is, a finite extension of $\mathbb{Q}$, and let $\mathcal{O}_F$ be its ring of integers.

Key object: The class group $\mathrm{Cl}(F) := I(F)/P(F)$ measures how far $\mathcal{O}_F$ is from being a principal ideal domain, or equivalently a unique factorization domain.

I. Iwasawa theory of class groups: number fields and class groups

Let F be a number field, that is, a finite extension of $\mathbb{Q}$, and let $\mathcal{O}_F$ be its ring of integers.

Key object: The class group $\mathrm{Cl}(F) := I(F)/P(F)$ measures how far $\mathcal{O}_F$ is from being a principal ideal domain, or equivalently a unique factorization domain.

Two examples

1. $\mathrm{Cl}(\mathbb{Q}) = \{1\}$, since $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$ is a principal ideal domain.

I. Iwasawa theory of class groups: number fields and class groups

Let F be a number field, that is, a finite extension of $\mathbb{Q}$, and let $\mathcal{O}_F$ be its ring of integers.

Key object: The class group $\mathrm{Cl}(F) := I(F)/P(F)$ measures how far $\mathcal{O}_F$ is from being a principal ideal domain, or equivalently a unique factorization domain.

Two examples

1. $\mathrm{Cl}(\mathbb{Q}) = \{1\}$, since $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$ is a principal ideal domain.
2. $\mathrm{Cl}(\mathbb{Q}(\sqrt{-5})) = \mathbb{Z}/2\mathbb{Z}$. In fact, $\mathcal{O}_{\mathbb{Q}(\sqrt{-5})} = \mathbb{Z}[\sqrt{-5}]$, and

$$6 = 2 \times 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}).$$

Theorem

The class group $\mathrm{Cl}(F)$ is a finite abelian group. We call $h(F) := \#\mathrm{Cl}(F)$ the class number of F.

I. Iwasawa theory of class groups: Gauss's problem

Gauss's problem

For quadratic fields $F = \mathbb{Q}(\sqrt{D})$ over $\mathbb{Q}$, how is $h(D) := \#\mathrm{Cl}(\mathbb{Q}(\sqrt{D}))$ distributed?

I. Iwasawa theory of class groups: Gauss's problem

Gauss's problem

For quadratic fields $F = \mathbb{Q}(\sqrt{D})$ over $\mathbb{Q}$, how is $h(D) := \#\mathrm{Cl}(\mathbb{Q}(\sqrt{D}))$ distributed?

Theorem (Heilbronn, 1934)

$h(D) \to \infty$, *as* $D \to -\infty$.

I. Iwasawa theory of class groups: Gauss's problem

Gauss's problem

For quadratic fields $F = \mathbb{Q}(\sqrt{D})$ over $\mathbb{Q}$, how is $h(D) := \#\mathrm{Cl}(\mathbb{Q}(\sqrt{D}))$ distributed?

Theorem (Heilbronn, 1934)

$h(D) \to \infty$, *as* $D \to -\infty$.

Theorem (Stark, 1966. Baker, 1971)

The only ***imaginary*** *quadratic fields of class number one are* $\mathbb{Q}(\sqrt{-D})$ *with*

$$D = 1, 2, 3, 7, 11, 19, 43, 67, 163.$$

I. Iwasawa theory of class groups: Gauss's problem

Gauss's problem

For quadratic fields $F = \mathbb{Q}(\sqrt{D})$ over $\mathbb{Q}$, how is $h(D) := \#\mathrm{Cl}(\mathbb{Q}(\sqrt{D}))$ distributed?

Theorem (Heilbronn, 1934)

$h(D) \to \infty$, as $D \to -\infty$.

Theorem (Stark, 1966. Baker, 1971)

*The only **imaginary** quadratic fields of class number one are $\mathbb{Q}(\sqrt{-D})$ with*

$$D = 1, 2, 3, 7, 11, 19, 43, 67, 163.$$

Remark: The parallel question for *real* quadratic fields is widely open.

I. Iwasawa theory of class groups: Horizontal and vertical families

Gauss's problem

Study class numbers in a **horizontal family**, for example all number fields F with $[F:\mathbb{Q}]=2$.

I. Iwasawa theory of class groups: Horizontal and vertical families

Gauss's problem

Study class numbers in a **horizontal family**, for example all number fields F with $[F:\mathbb{Q}]=2$.

Iwasawa's problem

Study class numbers in **vertical families** of number fields.

- An algebraic Galois extension F_∞/F is called a $\mathbb{Z}_p$-**extension** if $\mathrm{Gal}(F_\infty/F)\simeq\mathbb{Z}_p$.

I. Iwasawa theory of class groups: Horizontal and vertical families

Gauss's problem

Study class numbers in a **horizontal family**, for example all number fields F with $[F:\mathbb{Q}]=2$.

Iwasawa's problem

Study class numbers in **vertical families** of number fields.

- An algebraic Galois extension F_∞/F is called a $\mathbb{Z}_p$**-extension** if $\mathrm{Gal}(F_\infty/F)\simeq\mathbb{Z}_p$.
- The intermediate field structure is simple:

$$F\subset F_1\subseteq F_2\subset\cdots\subset F_n\subset\cdots\subset F_\infty,$$

where $\mathrm{Gal}(F_n/F)\simeq\mathbb{Z}/p^n$ for $n\geq 1$.

I. Iwasawa theory of class groups: examples – cyclotomic $\mathbb{Z}_p$-extensions

Let $p \geq 3$. Adjoining all p-power roots of unity gives

$$\mathrm{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}) \simeq \varprojlim_n \mathrm{Gal}(\mathbb{Q}(\mu_{p^n})/\mathbb{Q}) \simeq \varprojlim_n (\mathbb{Z}/p^n\mathbb{Z})^\times \simeq \mathbb{Z}_p^\times \simeq \mathbb{Z}/(p-1) \times (1+p\mathbb{Z}_p).$$

The subfield $\mathbb{Q}_\infty/\mathbb{Q}$ fixed by $\mathbb{Z}/(p-1)$ is a $\mathbb{Z}_p$-extension of $\mathbb{Q}$, called the **cyclotomic $\mathbb{Z}_p$-extension** of $\mathbb{Q}$.

[**Remark**: $1+p\mathbb{Z}_p$ is isomorphic to $\mathbb{Z}_p$ via the logarithm map, when $p \geq 3$.]

I. Iwasawa theory of class groups: examples – cyclotomic $\mathbb{Z}_p$-extensions

Let $p \geq 3$. Adjoining all p-power roots of unity gives

$$\mathrm{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}) \simeq \varprojlim_n \mathrm{Gal}(\mathbb{Q}(\mu_{p^n})/\mathbb{Q}) \simeq \varprojlim_n (\mathbb{Z}/p^n\mathbb{Z})^\times \simeq \mathbb{Z}_p^\times \simeq \mathbb{Z}/(p-1) \times (1+p\mathbb{Z}_p).$$

The subfield $\mathbb{Q}_\infty/\mathbb{Q}$ fixed by $\mathbb{Z}/(p-1)$ is a $\mathbb{Z}_p$-extension of $\mathbb{Q}$, called the **cyclotomic $\mathbb{Z}_p$-extension** of $\mathbb{Q}$.

[**Remark**: $1+p\mathbb{Z}_p$ is isomorphic to $\mathbb{Z}_p$ via the logarithm map, when $p \geq 3$.]

Similarly,

- $\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}(\mu_p)$ is a $\mathbb{Z}_p$-extension whose n-th layer is $\mathbb{Q}(\mu_{p^{n+1}})$,

I. Iwasawa theory of class groups: examples – cyclotomic $\mathbb{Z}_p$-extensions

Let $p \geq 3$. Adjoining all p-power roots of unity gives

$$\mathrm{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}) \simeq \varprojlim_n \mathrm{Gal}(\mathbb{Q}(\mu_{p^n})/\mathbb{Q}) \simeq \varprojlim_n (\mathbb{Z}/p^n\mathbb{Z})^\times \simeq \mathbb{Z}_p^\times \simeq \mathbb{Z}/(p-1) \times (1+p\mathbb{Z}_p).$$

The subfield $\mathbb{Q}_\infty/\mathbb{Q}$ fixed by $\mathbb{Z}/(p-1)$ is a $\mathbb{Z}_p$-extension of $\mathbb{Q}$, called the **cyclotomic $\mathbb{Z}_p$-extension** of $\mathbb{Q}$.

[**Remark**: $1+p\mathbb{Z}_p$ is isomorphic to $\mathbb{Z}_p$ via the logarithm map, when $p \geq 3$.]

Similarly,

- $\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}(\mu_p)$ is a $\mathbb{Z}_p$-extension whose n-th layer is $\mathbb{Q}(\mu_{p^{n+1}})$,
- $\mathbb{Q}(\mu_{p^\infty})^+/\mathbb{Q}(\mu_p)^+$ is a $\mathbb{Z}_p$-extension whose n-th layer is $\mathbb{Q}(\mu_{p^{n+1}})^+$. [Here F^+ denotes the maximal totally real subfield of F].

I. Iwasawa theory of class groups

Iwasawa's problem

Let F_∞/F be a $\mathbb{Z}_p$-**extension** of F with intermediate fields

$$F \subset F_1 \subseteq F_2 \subset \cdots \subset F_n \subset \cdots \subset F_\infty$$

Then how does the sequence $\{\#\mathrm{Cl}(F_n)[p^\infty]\}$ grow?

I. Iwasawa theory of class groups

Iwasawa's problem

Let F_∞/F be a $\mathbb{Z}_p$-**extension** of F with intermediate fields

$$F \subset F_1 \subseteq F_2 \subset \cdots \subset F_n \subset \cdots \subset F_\infty$$

Then how does the sequence $\{\#\mathrm{Cl}(F_n)[p^\infty]\}$ grow?

The key to the problem: consider all these class groups in one single giant object

$$\mathrm{Cl}_\infty := \varprojlim_n \mathrm{Cl}(F_n)[p^\infty].$$

where the transition maps are the norm maps

$$\mathrm{Nm}_{F_{n+1}/F_n} : \mathrm{Cl}(F_{n+1})[p^\infty] \longrightarrow \mathrm{Cl}(F_n)[p^\infty].$$

I. Iwasawa theory of class groups

The key to the problem: consider all these class groups in one single giant object

$$\mathrm{Cl}_\infty := \varprojlim_n \mathrm{Cl}(F_n)[p^\infty], \quad \mathrm{Nm}_{F_{n+1}/F_n} : \mathrm{Cl}(F_{n+1})[p^\infty] \longrightarrow \mathrm{Cl}(F_n)[p^\infty],$$

and each $\mathrm{Cl}(F_n)[p^\infty]$ carries an action of $\mathbb{Z}_p[\mathrm{Gal}(F_n/F)]$.

I. Iwasawa theory of class groups

The key to the problem: consider all these class groups in one single giant object

$$\mathrm{Cl}_\infty := \varprojlim_n \mathrm{Cl}(F_n)[p^\infty], \quad \mathrm{Nm}_{F_{n+1}/F_n} : \mathrm{Cl}(F_{n+1})[p^\infty] \longrightarrow \mathrm{Cl}(F_n)[p^\infty],$$

and each $\mathrm{Cl}(F_n)[p^\infty]$ carries an action of $\mathbb{Z}_p[\mathrm{Gal}(F_n/F)]$.

Advantage: Since the norm maps are Galois-equivariant, these actions are compatible. Consequently, Cl_∞ is naturally a module over

$$\Lambda := \varprojlim_n \mathbb{Z}_p[\mathrm{Gal}(F_n/F)] = \mathbb{Z}_p[\![\mathrm{Gal}(F_\infty/F)]\!] \simeq \mathbb{Z}_p[\![T]\!].$$

The key to the problem: consider all these class groups in one single giant object

$$\mathrm{Cl}_\infty := \varprojlim_n \mathrm{Cl}(F_n)[p^\infty], \quad \mathrm{Nm}_{F_{n+1}/F_n} : \mathrm{Cl}(F_{n+1})[p^\infty] \longrightarrow \mathrm{Cl}(F_n)[p^\infty],$$

and each $\mathrm{Cl}(F_n)[p^\infty]$ carries an action of $\mathbb{Z}_p[\mathrm{Gal}(F_n/F)]$.

Advantage: Since the norm maps are Galois-equivariant, these actions are compatible. Consequently, Cl_∞ is naturally a module over

$$\Lambda := \varprojlim_n \mathbb{Z}_p[\mathrm{Gal}(F_n/F)] = \mathbb{Z}_p[\![\mathrm{Gal}(F_\infty/F)]\!] \simeq \mathbb{Z}_p[\![T]\!].$$

The upshot: Cl_∞ is a **finitely generated torsion** Λ-module.

I. Iwasawa theory of class groups

The key to the problem: consider all these class groups in one single giant object

$$\mathrm{Cl}_\infty := \varprojlim_n \mathrm{Cl}(F_n)[p^\infty], \quad \mathrm{Nm}_{F_{n+1}/F_n} : \mathrm{Cl}(F_{n+1})[p^\infty] \longrightarrow \mathrm{Cl}(F_n)[p^\infty],$$

and each $\mathrm{Cl}(F_n)[p^\infty]$ carries an action of $\mathbb{Z}_p[\mathrm{Gal}(F_n/F)]$.

Advantage: Since the norm maps are Galois-equivariant, these actions are compatible. Consequently, Cl_∞ is naturally a module over

$$\Lambda := \varprojlim_n \mathbb{Z}_p[\mathrm{Gal}(F_n/F)] = \mathbb{Z}_p[\![\mathrm{Gal}(F_\infty/F)]\!] \simeq \mathbb{Z}_p[\![T]\!].$$

The upshot: Cl_∞ is a **finitely generated torsion** Λ-module.

Analogue: Each class group $\mathrm{Cl}(F_n)$ is a **finitely generated torsion** $\mathbb{Z}$-module.

I. Iwasawa theory of class groups

Here comes some commutative algebras of Λ-modules:

Finitely generated torsion Λ-modules are "rigid":

Let M be a finitely generated torsion Λ-module. Then there is a ("unique") map

$$M \to \bigoplus_{i=1}^{r} \Lambda/p^{m_i} \oplus \bigoplus_{j=1}^{t} \Lambda/f_j(T)^{n_j}$$

whose kernel and cokernel are of finite cardinality. Here $f_j(T) \in \mathbb{Z}_p[T]$ are "distinguished polynomials".

I. Iwasawa theory of class groups

Here comes some commutative algebras of Λ-modules:

Finitely generated torsion Λ-modules are "rigid":

Let M be a finitely generated torsion Λ-module. Then there is a ("unique") map

$$M \to \bigoplus_{i=1}^{r} \Lambda/p^{m_i} \oplus \bigoplus_{j=1}^{t} \Lambda/f_j(T)^{n_j}$$

whose kernel and cokernel are of finite cardinality. Here $f_j(T) \in \mathbb{Z}_p[T]$ are "distinguished polynomials". We can then define the Iwasawa-theoretic invariants of M:

- μ-**invariant**: $\mu(M) := \sum_{i=1}^{r} m_i$,

I. Iwasawa theory of class groups

Here comes some commutative algebras of Λ-modules:

Finitely generated torsion Λ-modules are "rigid":

Let M be a finitely generated torsion Λ-module. Then there is a ("unique") map

$$M \to \bigoplus_{i=1}^{r} \Lambda/p^{m_i} \oplus \bigoplus_{j=1}^{t} \Lambda/f_j(T)^{n_j}$$

whose kernel and cokernel are of finite cardinality. Here $f_j(T) \in \mathbb{Z}_p[T]$ are "distinguished polynomials". We can then define the Iwasawa-theoretic invariants of M:

- μ-**invariant**: $\mu(M) := \sum_{i=1}^{r} m_i$,
- λ-**invariant**: $\lambda(M) := \sum_{j=1}^{t} n_j \deg(f_j(T))$,

I. Iwasawa theory of class groups

Here comes some commutative algebras of Λ-modules:

Finitely generated torsion Λ-modules are "rigid":

Let M be a finitely generated torsion Λ-module. Then there is a ("unique") map

$$M \to \bigoplus_{i=1}^{r} \Lambda/p^{m_i} \oplus \bigoplus_{j=1}^{t} \Lambda/f_j(T)^{n_j}$$

whose kernel and cokernel are of finite cardinality. Here $f_j(T) \in \mathbb{Z}_p[T]$ are "distinguished polynomials". We can then define the Iwasawa-theoretic invariants of M:

- μ-**invariant**: $\mu(M) := \sum_{i=1}^{r} m_i$,
- λ-**invariant**: $\lambda(M) := \sum_{j=1}^{t} n_j \deg(f_j(T))$,
- **characteristic ideal**: $\mathrm{char}(M) := (p^{\mu(M)} \prod_{j=1}^{t} f_j(T)^{n_j})$, a principal ideal of Λ.

I. Iwasawa theory of class groups

Recall: The module Cl_∞ is a finitely generated torsion Λ-module. Using homological algebra, we can trace back to finite levels and obtain the following theorem.

Iwasawa's class number formula

$$\#\mathrm{Cl}(F_n)[p^\infty] = p^{\mu p^n + \lambda n + \nu}, \quad \text{for } n \gg 0,$$

where $\mu = \mu(\mathrm{Cl}_\infty)$, $\lambda = \lambda(\mathrm{Cl}_\infty)$, and ν is an integer (possibly negative).

Is this really a big result? Yes and No!

I. Iwasawa theory of class groups

Recall: The module Cl_∞ is a finitely generated torsion Λ-module. Using homological algebra, we can trace back to finite levels and obtain the following theorem.

Iwasawa's class number formula

$$\#\mathrm{Cl}(F_n)[p^\infty] = p^{\mu p^n + \lambda n + \nu}, \quad \text{for } n \gg 0,$$

where $\mu = \mu(\mathrm{Cl}_\infty)$, $\lambda = \lambda(\mathrm{Cl}_\infty)$, and ν is an integer (possibly negative).

Is this really a big result? Yes and No!

- **Yes**: At least we now understand more about the growth pattern of $\mathrm{Cl}(F_n)[p^\infty]$.

I. Iwasawa theory of class groups

Recall: The module Cl_∞ is a finitely generated torsion Λ-module. Using homological algebra, we can trace back to finite levels and obtain the following theorem.

Iwasawa's class number formula

$$\#\mathrm{Cl}(F_n)[p^\infty] = p^{\mu p^n + \lambda n + \nu}, \quad \text{for } n \gg 0,$$

where $\mu = \mu(\mathrm{Cl}_\infty)$, $\lambda = \lambda(\mathrm{Cl}_\infty)$, and ν is an integer (possibly negative).

Is this really a big result? Yes and No!

- **Yes**: At least we now understand more about the growth pattern of $\mathrm{Cl}(F_n)[p^\infty]$.
- **No**: "These are merely algebraic tricks; we still do not know the invariants μ, λ, and ν."

I. Iwasawa theory of class groups

Recall: The module Cl_∞ is a finitely generated torsion Λ-module. Using homological algebra, we can trace back to finite levels and obtain the following theorem.

Iwasawa's class number formula

$$\#\mathrm{Cl}(F_n)[p^\infty] = p^{\mu p^n + \lambda n + \nu}, \quad \text{for } n \gg 0,$$

where $\mu = \mu(\mathrm{Cl}_\infty)$, $\lambda = \lambda(\mathrm{Cl}_\infty)$, and ν is an integer (possibly negative).

Is this really a big result? Yes and No!

- **Yes**: At least we now understand more about the growth pattern of $\mathrm{Cl}(F_n)[p^\infty]$.
- **No**: "These are merely algebraic tricks; we still do not know the invariants μ, λ, and ν."

The upshot: The goal is to describe the invariants $\mu(\mathrm{Cl}_\infty)$, $\lambda(\mathrm{Cl}_\infty)$, and even the characteristic ideal $\mathrm{char}(\mathrm{Cl}_\infty)$ **analytically**.

Four Themes of Iwasawa Theory (Imprecise):

Let $F_\infty/F = \mathbb{Q}(\mu_{p^\infty})^+/\mathbb{Q}(\mu_p)^+$.

(A) **Algebraic side**: Cl_∞ is a finitely generated torsion Λ-module.

Four Themes of Iwasawa Theory (Imprecise):

Let $F_\infty/F = \mathbb{Q}(\mu_{p^\infty})^+/\mathbb{Q}(\mu_p)^+$.

(A) **Algebraic side**: Cl_∞ is a finitely generated torsion Λ-module.

(B) **Analytic side** [Kubota-Leopoldt, 1960s]: There exist ***p*-adic *L*-functions** $\mathcal{L}_p \in \Lambda$ encoding the special values of Dirichlet L-functions:

$$\{L(\psi\omega^{-r}\phi^{-1}, 1-r) \mid r \geq 1,\ \phi : \mathrm{Gal}(F_\infty/F) \to \mathbb{C}^\times \text{ of finite order}\}.$$

Four Themes of Iwasawa Theory (Imprecise):

Let $F_\infty/F = \mathbb{Q}(\mu_{p^\infty})^+/\mathbb{Q}(\mu_p)^+$.

(A) **Algebraic side**: Cl_∞ is a finitely generated torsion Λ-module.

(B) **Analytic side** [Kubota-Leopoldt, 1960s]: There exist ***p*-adic *L*-functions** $\mathcal{L}_p \in \Lambda$ encoding the special values of Dirichlet L-functions:

$$\{L(\psi\omega^{-r}\phi^{-1}, 1-r) \mid r \geq 1,\ \phi : \mathrm{Gal}(F_\infty/F) \to \mathbb{C}^\times \text{ of finite order}\}.$$

(C) **Iwasawa's Main Conjecture** [Mazur-Wiles, 1980s]: As principal ideals of Λ, we have the equality

$$\mathrm{char}(\mathrm{Cl}_\infty) = (\mathcal{L}_p).$$

Four Themes of Iwasawa Theory (Imprecise):

Let $F_\infty/F = \mathbb{Q}(\mu_{p^\infty})^+/\mathbb{Q}(\mu_p)^+$.

(A) **Algebraic side**: Cl_∞ is a finitely generated torsion Λ-module.

(B) **Analytic side** [Kubota-Leopoldt, 1960s]: There exist ***p*-adic *L*-functions** $\mathcal{L}_p \in \Lambda$ encoding the special values of Dirichlet L-functions:

$$\{L(\psi\omega^{-r}\phi^{-1}, 1-r) \mid r \geq 1,\ \phi : \mathrm{Gal}(F_\infty/F) \to \mathbb{C}^\times \text{ of finite order}\}.$$

(C) **Iwasawa's Main Conjecture** [Mazur-Wiles, 1980s]: As principal ideals of Λ, we have the equality

$$\mathrm{char}(\mathrm{Cl}_\infty) = (\mathcal{L}_p).$$

(D) $\mu(\mathcal{L}_p) = 0$ [Ferrero-Washington, 1980s], and $\lambda(\mathcal{L}_p)$ is *conjectured* to be zero [Greenberg, 1980s].

Four Themes of Iwasawa Theory (Imprecise):

Let $F_\infty/F = \mathbb{Q}(\mu_{p^\infty})^+/\mathbb{Q}(\mu_p)^+$.

(C) **Iwasawa's Main Conjecture** [Mazur-Wiles, 1980s]: As principal ideals of Λ, we have the equality

$$\mathrm{char}(\mathrm{Cl}_\infty) = (\mathcal{L}_p).$$

(D) $\mu(\mathcal{L}_p) = 0$ [Ferrero-Washington, 1980s], and $\lambda(\mathcal{L}_p)$ is *conjectured* to be zero [Greenberg, 1980s].

Corollary: Granting Greenberg's conjecture, $\#\mathrm{Cl}(\mathbb{Q}(\mu_{p^n})^+)[p^\infty]$ is bounded, and moreover **eventually constant** as $n \to \infty$.

Four Themes of Iwasawa Theory (Imprecise):

Let $F_\infty/F = \mathbb{Q}(\mu_{p^\infty})^+/\mathbb{Q}(\mu_p)^+$.

(C) **Iwasawa's Main Conjecture** [Mazur-Wiles, 1980s]: As principal ideals of Λ, we have the equality

$$\mathrm{char}(\mathrm{Cl}_\infty) = (\mathcal{L}_p).$$

(D) $\mu(\mathcal{L}_p) = 0$ [Ferrero-Washington, 1980s], and $\lambda(\mathcal{L}_p)$ is *conjectured* to be zero [Greenberg, 1980s].

Corollary: Granting Greenberg's conjecture, $\#\mathrm{Cl}(\mathbb{Q}(\mu_{p^n})^+)[p^\infty]$ is bounded, and moreover **eventually constant** as $n \to \infty$.

Example: Iwasawa constructed examples of **non-cyclotomic** $\mathbb{Z}_3$-extensions F_∞^{ac} of $F := \mathbb{Q}(\mu_3, \sqrt[3]{10}) = \mathbb{Q}(\sqrt{-3}, \sqrt[3]{10})$ such that $\mu(\mathrm{Cl}_\infty) > 0$. In this case, the class number $\mathrm{Cl}(F_n^{\mathrm{ac}})[3^\infty]$ is unbounded. [Iwasawa's construction is far more general]

I. Iwasawa theory of class groups: Eisenstein congruences

Today, we focus on **Theme (C)**: Iwasawa's main conjecture:

$$\mathrm{char}(\mathrm{Cl}_\infty) = (\mathcal{L}_p),$$

Upper bound for Cl_∞: Show $\mathrm{char}(\mathrm{Cl}_\infty) \mid \mathcal{L}_p$

Use **Euler system** to give an upper bound of the class group Cl_∞.

I. Iwasawa theory of class groups: Eisenstein congruences

Today, we focus on **Theme (C)**: Iwasawa's main conjecture:

$$\mathrm{char}(\mathrm{Cl}_\infty) = (\mathcal{L}_p),$$

Upper bound for Cl_∞: Show $\mathrm{char}(\mathrm{Cl}_\infty) \mid \mathcal{L}_p$

Use **Euler system** to give an upper bound of the class group Cl_∞.

Our main focus today:

Lower bound for Cl_∞: Show $\mathcal{L}_p \mid \mathrm{char}(\mathrm{Cl}_\infty)$

Use **Eisenstein congruences** to construct sufficiently many ideal classes.

Baby example: Serre's construction with Ramanujan's congruence

Consider $p = 691$. How to construct a nontrivial p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Baby example: Serre's construction with Ramanujan's congruence

Consider $p = 691$. How to construct a nontrivial p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Step 1 : Consider the weight 12, level $\mathrm{SL}_2(\mathbb{Z})$ Eisenstein series

$$E_{12}(q = \exp(2\pi \mathrm{i} z)) = -\frac{B_{12}}{24} + \sum_{n \geq 1} \left(\sum_{d|n} d^{11} \right) q^n, \quad \frac{B_{12}}{24} = \frac{\boxed{691}}{156} \cdot \frac{1}{420}.$$

Baby example: Serre's construction with Ramanujan's congruence

Consider $p = 691$. How to construct a nontrivial p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Step 1 : Consider the weight 12, level $\mathrm{SL}_2(\mathbb{Z})$ Eisenstein series

$$E_{12}(q = \exp(2\pi \mathrm{i} z)) = -\frac{B_{12}}{24} + \sum_{n \geq 1} \left(\sum_{d|n} d^{11} \right) q^n, \quad \frac{B_{12}}{24} = \frac{\boxed{691}}{156} \cdot \frac{1}{420}.$$

Step 2 : Therefore, p divides the constant term of E_{12}. Therefore it is plausible to construct a cuspform $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ such that $E_{12} \equiv f \pmod{p}$.

Baby example: Serre's construction with Ramanujan's congruence

Consider $p = 691$. How to construct a nontrivial p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Step 1 : Consider the weight 12, level $\mathrm{SL}_2(\mathbb{Z})$ Eisenstein series

$$E_{12}(q = \exp(2\pi \mathrm{i} z)) = -\frac{B_{12}}{24} + \sum_{n \geq 1} \left(\sum_{d|n} d^{11} \right) q^n, \quad \frac{B_{12}}{24} = \boxed{\frac{691}{156}} \cdot \frac{1}{420}.$$

Step 2 : Therefore, p divides the constant term of E_{12}. Therefore it is plausible to construct a cuspform $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ such that $E_{12} \equiv f \pmod{p}$. Actually it is proved by Ramanujan that

$$\Delta \equiv E_{12} \mod p = 691,$$

where

$$\Delta(q = \exp(2\pi \mathrm{i} z)) := q \prod_{i=1}^{\infty} (1 - q^n)^{24} = \sum_{n=1}^{\infty} \tau(n) q^n \in S_{12}(\mathrm{SL}_2(\mathbb{Z})), \quad \mathrm{im}(z) > 0.$$

Baby example: Serre's construction with Ramanujan's congruence

Consider $p = 691$. How to construct a nontrivial p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Step 2 : Ramanujan's congruence

$$\Delta \equiv E_{12} \mod p = 691.$$

Baby example: Serre's construction with Ramanujan's congruence

Consider $p = 691$. How to construct a nontrivial p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Step 2 : Ramanujan's congruence

$$\Delta \equiv E_{12} \mod p = 691.$$

Step 3 : Translate this to a modulo-p congruence of Galois representations, we have

$$\rho_\Delta \equiv \rho_{E_{12}} \equiv \begin{bmatrix} \omega^{11} & \star \\ 0 & 1 \end{bmatrix} \pmod{691}.$$

The Galois cohomology class $\star$ provides a p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$.

Baby example: Serre's construction with Ramanujan's congruence

Consider $p = 691$. How to construct a nontrivial p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Step 3 : We have

$$\rho_\Delta \equiv \rho_{E_{12}} \equiv \begin{bmatrix} \omega^{11} & \star \\ 0 & 1 \end{bmatrix} \pmod{691}.$$

The Galois cohomology class $\star$ provides a p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$.

p-primitivity of the Eisenstein series E_{12}

Question: How to show that $\star$ gives a **nontrivial** ideal class in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Baby example: Serre's construction with Ramanujan's congruence

Consider $p = 691$. How to construct a nontrivial p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Step 3 : We have

$$\rho_\Delta \equiv \rho_{E_{12}} \equiv \begin{bmatrix} \omega^{11} & \star \\ 0 & 1 \end{bmatrix} \pmod{691}.$$

The Galois cohomology class $\star$ provides a p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$.

p-primitivity of the Eisenstein series E_{12}

Question: How to show that $\star$ gives a **nontrivial** ideal class in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Question: Is the congruence $\Delta \equiv E_{12} \pmod{p = 691}$ nontrivial?

Baby example: Serre's construction with Ramanujan's congruence

Consider $p = 691$. How to construct a nontrivial p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Step 3 : We have

$$\rho_\Delta \equiv \rho_{E_{12}} \equiv \begin{bmatrix} \omega^{11} & \star \\ 0 & 1 \end{bmatrix} \pmod{691}.$$

The Galois cohomology class $\star$ provides a p-torsion element in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$.

p-primitivity of the Eisenstein series E_{12}

Question: How to show that $\star$ gives a **nontrivial** ideal class in $\mathrm{Cl}(\mathbb{Q}(\mu_p))$?

Question: Is the congruence $\Delta \equiv E_{12} \pmod{p = 691}$ nontrivial?

Answer: The p-th Fourier coefficient of E_{12} is $1 + p^{11}$, which is nonzero modulo p. Hence $E_{12} \not\equiv 0 \pmod{p}$. – i.e. E_{12} is p-**primitive**.

Inspired by Ramanujan's congruence, **B. Mazur and A. Wiles first applied to Iwasawa theory** the idea of constructing congruences modulo p between cuspidal eigenforms and Eisenstein series, namely, **Eisenstein congruences** ([Mazur–Wiles, '84] [Wiles, '90]), to prove the Iwasawa main conjecture for GL_1 over totally real fields, that is, for class groups.

- B. Mazur and A. Wiles. *Class fields of abelian extensions of* $\mathbb{Q}$. Invent. Math., 1984.
- A. Wiles. *The Iwasawa conjecture for totally real fields*. Annals of Mathematics, 1990.

II. Iwasawa theory for elliptic curves – Step to unitary groups

II. Iwasawa theory for elliptic curves

Mazur, Greenberg, ...: Iwasawa theory of class groups can be vastly generalized to study the **arithmetic of elliptic curves**, modular forms, and even general Galois representations.

Question: Iwasawa theory of elliptic curves

Suppose we want to study the Iwasawa theory of an elliptic curve $A/\mathbb{Q}$ [with good ordinary reduction at p] . How does the machinery of Eisenstein congruences proceed?

II. Iwasawa theory for elliptic curves

Baby Case Step 1 : Regard the Dirichlet L-function $L(\chi, s)$ of a Dirichlet character χ as an automorphic L-function over the reductive group GL_1, and construct an Eisenstein series over GL_2 whose constrant term involves $L(\chi, s)$.

II. Iwasawa theory for elliptic curves

Baby Case Step 1 : Regard the Dirichlet L-function $L(\chi, s)$ of a Dirichlet character χ as an automorphic L-function over the reductive group GL_1, and construct an Eisenstein series over GL_2 whose constrant term involves $L(\chi, s)$.

Step 1 now : Regard the Hasse–Weil L-function $L(A, s)$ of the elliptic curve $A/\mathbb{Q}$ as an automorphic L-function over the reductive group GL_2, the task is to construct

an Eisenstein series E_A over certain reductive group $\mathbf{G}$ (?) contaning GL_2

whose constant term should involve $L(A, s)$.

II. Iwasawa theory of unitary groups

Upshot (Langlands–Shahidi): The **Klingen Eisenstein series** E_A^{Kling} over unitary groups $\mathrm{U}(2,2)$ or $\mathrm{U}(3,1)$ encodes $L(A,s)$.

II. Iwasawa theory of unitary groups

Upshot (Langlands–Shahidi): The **Klingen Eisenstein series** E_A^{Kling} over unitary groups $\mathrm{U}(2,2)$ or $\mathrm{U}(3,1)$ encodes $L(A,s)$.

Unitary groups

Let K be an auxiliary imaginary quadratic field such that p splits in K, one defines

$$\mathrm{U}(m,n) := \left\{ g \in \mathrm{GL}_{m+n}(K) : \overline{g}^{\mathrm{t}} \begin{bmatrix} \mathbf{1}_m & 0 \\ 0 & -\mathbf{1}_n \end{bmatrix} g = \begin{bmatrix} \mathbf{1}_m & 0 \\ 0 & -\mathbf{1}_n \end{bmatrix} \right\}.$$

This is called the **unitary group** of signature (m,n). This is an algebraic group over $\mathbb{Q}$.

II. Iwasawa theory of unitary groups

Upshot (Langlands–Shahidi): The **Klingen Eisenstein series** E_A^{Kling} over unitary groups $\mathrm{U}(2,2)$ or $\mathrm{U}(3,1)$ encodes $L(A,s)$.

Unitary groups

Let K be an auxiliary imaginary quadratic field such that p splits in K, one defines

$$\mathrm{U}(m,n) := \left\{ g \in \mathrm{GL}_{m+n}(K) : \overline{g}^{\mathrm{t}} \begin{bmatrix} \mathbf{1}_m & 0 \\ 0 & -\mathbf{1}_n \end{bmatrix} g = \begin{bmatrix} \mathbf{1}_m & 0 \\ 0 & -\mathbf{1}_n \end{bmatrix} \right\}.$$

This is called the **unitary group** of signature (m,n). This is an algebraic group over $\mathbb{Q}$. For example,

- $\mathrm{U}(1,1) \simeq \mathrm{GL}_2$, and $\mathrm{U}(1,1) \hookrightarrow \mathrm{U}(2,2)$,
- $\mathrm{U}(2,0) \simeq$ a nonsplit quaternion algebra, and $\mathrm{U}(2,0) \hookrightarrow \mathrm{U}(3,1)$.

Let's put ourselves in the most general case:

Step 1 [Langlands–Shahidi] Let Φ be a cuspidal automorphic form over the unitary group $\mathrm{U}(m,n)$, we can define a Klingen Eisenstein series

$$E_{\Phi}^{\mathrm{Kling}} : \mathrm{U}(m+1,n+1)(\mathbb{A}_{\mathbb{Q}}) \to \mathbb{C}$$

with constant term essentially the L-function of Φ.

Let's put ourselves in the most general case:

Step 1 [Langlands–Shahidi] Let Φ be a cuspidal automorphic form over the unitary group $\mathrm{U}(m,n)$, we can define a Klingen Eisenstein series

$$E_\Phi^{\mathrm{Kling}} : \mathrm{U}(m+1,n+1)(\mathbb{A}_\mathbb{Q}) \to \mathbb{C}$$

with constant term essentially the L-function of Φ.

Step 2 [Hida, ...] Suppose p divides (the algebraic part of) $L(\Phi, s)$, then it leads to the expectation that "$E_\Phi^{\mathrm{Kling}} \equiv f \pmod{p}$" for some cuspidal eigenform f over $\mathrm{U}(m+1,n+1)$.

Let's put ourselves in the most general case:

Step 1 [Langlands–Shahidi] Let Φ be a cuspidal automorphic form over the unitary group $\mathrm{U}(m,n)$, we can define a Klingen Eisenstein series

$$E_\Phi^{\mathrm{Kling}} : \mathrm{U}(m+1,n+1)(\mathbb{A}_\mathbb{Q}) \to \mathbb{C}$$

with constant term essentially the L-function of Φ.

Step 2 [Hida, ...] Suppose p divides (the algebraic part of) $L(\Phi, s)$, then it leads to the expectation that "$E_\Phi^{\mathrm{Kling}} \equiv f \pmod{p}$" for some cuspidal eigenform f over $\mathrm{U}(m+1,n+1)$.

Let's put ourselves in the most general case:

Step 1 [Langlands–Shahidi] Let Φ be a cuspidal automorphic form over the unitary group $\mathrm{U}(m,n)$, we can define a Klingen Eisenstein series

$$E_\Phi^{\mathrm{Kling}} : \mathrm{U}(m+1,n+1)(\mathbb{A}_\mathbb{Q}) \to \mathbb{C}$$

with constant term essentially the L-function of Φ.

Step 2 [Hida, ...] Suppose p divides (the algebraic part of) $L(\Phi, s)$, then it leads to the expectation that "$E_\Phi^{\mathrm{Kling}} \equiv f \pmod{p}$" for some cuspidal eigenform f over $\mathrm{U}(m+1,n+1)$.

The key difficulty: "*Step 2*" of Eisenstein congruences

How to show that the congruence relation "$E_\Phi^{\mathrm{Kling}} \equiv f \pmod{p}$" is **nontrivial**? In other words, how to show that $E_\Phi^{\mathrm{Kling}} \not\equiv 0 \pmod{p}$? (after appropriate normalization to make it algebraic.)

The key difficulty: "*Step 2*" of Eisenstein congruences

How to show that the congruence relation "$E_{\Phi}^{\mathrm{Kling}} \equiv f \pmod{p}$" is **nontrivial**? In other words, how to show that $E_{\Phi}^{\mathrm{Kling}} \not\equiv 0 \pmod{p}$?

Example on lower rank cases

- GL_2-case: Compute its p-th Fourier coefficient of E_{12}, which is nonzero modulo p.

II. Iwasawa theory of unitary groups: Eisenstein congruences

The key difficulty: "*Step 2*" of Eisenstein congruences

How to show that the congruence relation "$E_{\Phi}^{\mathrm{Kling}} \equiv f \pmod{p}$" is **nontrivial**? In other words, how to show that $E_{\Phi}^{\mathrm{Kling}} \not\equiv 0 \pmod{p}$?

Example on lower rank cases

- GL_2-case: Compute its p-th Fourier coefficient of E_{12}, which is nonzero modulo p.
- For Klingen Eisenstein series E_A^{Kling} over $\mathrm{U}(2,2)$ or $\mathrm{U}(3,1)$, certain Fourier coefficients of E_A^{Kling} can be computed.

  [Related works: [Skinner-Urban 2014], [Wan 2015], [Wan 2020],[Castella-Liu-Wan 2022]].

II. Iwasawa theory of unitary groups: Eisenstein congruences

The key difficulty: "*Step 2*" of Eisenstein congruences

How to show that the congruence relation "$E_{\Phi}^{\mathrm{Kling}} \equiv f \pmod{p}$" is **nontrivial**? In other words, how to show that $E_{\Phi}^{\mathrm{Kling}} \not\equiv 0 \pmod{p}$?

Example on lower rank cases

- GL_2-case: Compute its p-th Fourier coefficient of E_{12}, which is nonzero modulo p.
- For Klingen Eisenstein series E_A^{Kling} over $\mathrm{U}(2,2)$ or $\mathrm{U}(3,1)$, certain Fourier coefficients of E_A^{Kling} can be computed.

  [Related works: [Skinner-Urban 2014], [Wan 2015], [Wan 2020],[Castella-Liu-Wan 2022]].

Problem: these methods are hard to generalize to higher ranks.

The key difficulty: "*Step 2*" of Eisenstein congruences

How to show that the congruence relation "$E_{\Phi}^{\mathrm{Kling}} \equiv f \pmod{p}$" is **nontrivial**? In other words, how to show that $E_{\Phi}^{\mathrm{Kling}} \not\equiv 0 \pmod{p}$?

- **Previous work**: Compute Fourier coefficients of E_{A}^{Kling}. This applies only in low-rank cases.
- **New method**: Compute Gan–Gross–Prasad type period integrals of $E_{\Phi}^{\mathrm{Kling}}$ on unitary groups of arbitrary rank.

II. GGP periods of Klingen Eisenstein series: general setup

- Let $\mathcal{K}/\mathcal{F}$ be a CM extension, so $\mathcal{F}$ is totally real and $\mathcal{K}$ is a totally imaginary quadratic extension of $\mathcal{F}$. (Previously we are using an imaginary quadratic field $\mathcal{K}/\mathbb{Q}$.)
- Let $p > 2$ be unramified in $\mathcal{F}$, and assume that every place of $\mathcal{F}$ above p splits in $\mathcal{K}$.

II. GGP periods of Klingen Eisenstein series: general setup

- Let $\mathcal{K}/\mathcal{F}$ be a CM extension, so $\mathcal{F}$ is totally real and $\mathcal{K}$ is a totally imaginary quadratic extension of $\mathcal{F}$. (Previously we are using an imaginary quadratic field $\mathcal{K}/\mathbb{Q}$.)
- Let $p > 2$ be unramified in $\mathcal{F}$, and assume that every place of $\mathcal{F}$ above p splits in $\mathcal{K}$.
- Let $m \geq n$. We write $\mathrm{U}(m,n)$ for the unitary group over $\mathcal{F}$ of signature (m,n):

$$\mathrm{U}(m,n)(\mathcal{F}) := \left\{ g \in \mathrm{GL}_{m+n}(\mathcal{K}) : \overline{g}^{t} \begin{bmatrix} \mathbf{1}_m & 0 \\ 0 & -\mathbf{1}_n \end{bmatrix} g = \begin{bmatrix} \mathbf{1}_m & 0 \\ 0 & -\mathbf{1}_n \end{bmatrix} \right\}.$$

II. GGP periods of Klingen Eisenstein series: general setup

Main objects

- Let σ be an irreducible, algebraic, regular, unitary, cuspidal representation of $H := \mathrm{U}(m,n)$, with σ_∞ in the holomorphic discrete series.

$\rightsquigarrow$ Let Φ be a cuspidal automorphic form in σ.

- Let $\chi : \mathcal{K}^\times \backslash \mathbb{A}_{\mathcal{K}}^\times \to \mathbb{C}^\times$ be a Hecke character of $\mathcal{K}$.

To study the Iwasawa theory of σ, more precisely of $\sigma \boxtimes \chi$:

II. GGP periods of Klingen Eisenstein series: general setup

Main objects

- Let σ be an irreducible, algebraic, regular, unitary, cuspidal representation of $H := \mathrm{U}(m,n)$, with σ_∞ in the holomorphic discrete series.

$\rightsquigarrow$ Let Φ be a cuspidal automorphic form in σ.

- Let $\chi : \mathcal{K}^\times \backslash \mathbb{A}_{\mathcal{K}}^\times \to \mathbb{C}^\times$ be a Hecke character of $\mathcal{K}$.

To study the Iwasawa theory of σ, more precisely of $\sigma \boxtimes \chi$:

We study the **Klingen Eisenstein series** $E_{\sigma,\chi}^{\mathrm{Kling}}$ on $G^\heartsuit := \mathrm{U}(m+1, n+1)$.

In particular, is it p-**primitive, meaning nonzero modulo** p?

II. GGP periods of Klingen Eisenstein series: definition

$$H := \mathrm{U}(m,n) \overset{\jmath^\sharp}{\hookrightarrow} G := \mathrm{U}(m+1,n) \overset{\jmath^\flat}{\hookrightarrow} G^\heartsuit = \mathrm{U}(m+1,n+1)$$

We use the following auxiliary input:

- (π, V_π) is an irreducible unitary cuspidal automorphic representation of G over $\mathcal{F}$, and we regard V_π as a subspace of $\mathcal{A}_{\mathrm{cusp}}(G(\mathcal{F})\backslash G(\mathbb{A}_{\mathcal{F}}))$.
- $\Psi \in V_\pi$.

II. GGP periods of Klingen Eisenstein series: definition

$$H := \mathrm{U}(m,n) \overset{\jmath^\sharp}{\hookrightarrow} G := \mathrm{U}(m+1,n) \overset{\jmath^\flat}{\hookrightarrow} G^\heartsuit = \mathrm{U}(m+1,n+1)$$

We use the following auxiliary input:

- (π, V_π) is an irreducible unitary cuspidal automorphic representation of G over $\mathcal{F}$, and we regard V_π as a subspace of $\mathcal{A}_{\mathrm{cusp}}(G(\mathcal{F})\backslash G(\mathbb{A}_\mathcal{F}))$.
- $\Psi \in V_\pi$.

Definition

For $\Psi \in V_\pi$, we define the *Gan–Gross–Prasad period integral* of $E^{\mathrm{Kling}}_{\sigma,\chi}$ against the cusp form Ψ by

$$\mathcal{P}(\Psi, E^{\mathrm{Kling}}_{\sigma,\chi}) := \int_{G(\mathcal{F})\backslash G(\mathbb{A}_\mathcal{F})} E^{\mathrm{Kling}}_{\sigma,\chi}(\jmath^\flat(g))\Psi(g)\,\mathrm{d}g.$$

II. GGP periods of Klingen Eisenstein series: definition

$$H := \mathrm{U}(m,n) \overset{\jmath^\sharp}{\hookrightarrow} G := \mathrm{U}(m+1,n) \overset{\jmath^\flat}{\hookrightarrow} G^\heartsuit = \mathrm{U}(m+1,n+1)$$

$$\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi} := \mathcal{P}(\Psi, E^{\mathrm{Kling}}_{\sigma,\chi}) := \int_{G(\mathcal{F})\backslash G(\mathbb{A}_{\mathcal{F}})} E^{\mathrm{Kling}}_{\sigma,\chi}(\jmath^\flat(g))\Psi(g)\,\mathrm{d}g.$$

Why consider this period integral?

- To prove that $E^{\mathrm{Kling}}_{\sigma,\chi}$ is nonzero modulo p, it suffices to **choose an appropriate** Ψ such that $\mathcal{P}(\Psi, E^{\mathrm{Kling}}_{\sigma,\chi})$ is nonzero modulo p.

II. GGP periods of Klingen Eisenstein series: definition

$$H := \mathrm{U}(m,n) \overset{\jmath^\sharp}{\hookrightarrow} G := \mathrm{U}(m+1,n) \overset{\jmath^\flat}{\hookrightarrow} G^\heartsuit = \mathrm{U}(m+1,n+1)$$

$$\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi} := \mathcal{P}(\Psi, E^{\mathrm{Kling}}_{\sigma,\chi}) := \int_{G(\mathcal{F})\backslash G(\mathbb{A}_{\mathcal{F}})} E^{\mathrm{Kling}}_{\sigma,\chi}(\jmath^\flat(g))\Psi(g)\,\mathrm{d}g.$$

Why consider this period integral?

- To prove that $E^{\mathrm{Kling}}_{\sigma,\chi}$ is nonzero modulo p, it suffices to **choose an appropriate** Ψ such that $\mathcal{P}(\Psi, E^{\mathrm{Kling}}_{\sigma,\chi})$ is nonzero modulo p.
- Such period integrals are closely related to **special values of** L**-functions**, for which more tools are available.

Main Theorem A: with the explicit construction of Klingen Eisenstein series in [Wan 15]

$$\frac{\left(\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi}\right)^2}{||\Phi||^2_{\sigma,\mathtt{MVW}}\,||\Psi||^2_{\pi,\mathtt{MVW}}} = \frac{1}{2^{\varkappa_\sigma+\varkappa_\pi}}\cdot \mathscr{L}_{\mathscr{S}^{\mathrm{ur}}_{\mathcal{F}}}(\sigma\times\pi) L_{\mathscr{U}^{\mathrm{ur}}_{\mathcal{F}}}\left(s+\frac{1}{2},\pi^\vee,\overline{\chi}^{\mathrm{c}}\right)^2$$
$$\times \prod_{v\in\mathscr{U}^{\mathrm{ur}}_{\mathcal{F}}} d_{N+1,v}(s,\overline{\chi}_v)^{-2} \prod_{v\in\mathscr{V}^{\mathrm{bad}}_{\mathcal{F}}} \mathscr{I}_v(\sigma_v,\pi_v) \prod_{v\in\mathscr{U}^{\mathrm{bad}}_{\mathcal{F}}} \mathscr{Z}_v^{\diamondsuit,\triangleright}(f^{\mathrm{Sieg}}_{v,s,\chi},\pi_v)^2,$$

For any set $\mathscr{S}$ of finite places of $\mathcal{F}$, the **local Rankin–Selberg L-factor** is

$$\mathscr{L}_{\mathscr{S}}(\sigma\times\pi) := \prod_{v\in\mathscr{S}}\left(\frac{L(\frac{1}{2},\sigma_v\times\pi_v)}{L(1,\sigma_v,\mathrm{Ad})L(1,\pi_v,\mathrm{Ad})}\prod_{i=1}^{m+n+1} L(i,\epsilon^i_{\mathcal{K}_v/\mathcal{F}_v})\right),$$

Main Theorem A: with the explicit construction of Klingen Eisenstein series in [Wan 15]

$$\frac{\left(\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi}\right)^2}{||\Phi||^2_{\sigma,\mathtt{MVW}}||\Psi||^2_{\pi,\mathtt{MVW}}} = \frac{1}{2^{\varkappa_\sigma+\varkappa_\pi}} \cdot \mathscr{L}_{\mathscr{S}^{\mathrm{ur}}_{\mathcal{F}}}(\sigma\times\pi) L_{\mathscr{U}^{\mathrm{ur}}_{\mathcal{F}}}\left(s+\frac{1}{2}, \pi^\vee, \overline{\chi}^{\mathrm{c}}\right)^2$$
$$\times \prod_{v\in\mathscr{U}^{\mathrm{ur}}_{\mathcal{F}}} d_{N+1,v}(s,\overline{\chi}_v)^{-2} \prod_{v\in\mathscr{V}^{\mathrm{bad}}_{\mathcal{F}}} \mathscr{I}_v(\sigma_v,\pi_v) \prod_{v\in\mathscr{U}^{\mathrm{bad}}_{\mathcal{F}}} \mathscr{Z}_v^{\diamondsuit,\triangleright}(f^{\mathrm{Sieg}}_{v,s,\chi},\pi_v)^2,$$

For any set $\mathscr{S}$ of finite places of $\mathcal{F}$, the **local Rankin–Selberg L-factor** is

$$\mathscr{L}_{\mathscr{S}}(\sigma\times\pi) := \prod_{v\in\mathscr{S}} \left(\frac{L(\frac{1}{2},\sigma_v\times\pi_v)}{L(1,\sigma_v,\mathrm{Ad})L(1,\pi_v,\mathrm{Ad})} \prod_{i=1}^{m+n+1} L(i,\epsilon^i_{\mathcal{K}_v/\mathcal{F}_v}) \right),$$

and the **local doubling-method L-factor** is

$$L_{\mathscr{S}}(s,\pi,\chi) := \prod_{v\in\mathscr{S}} L_v(s,\mathrm{BC}(\pi_v)\otimes\chi_v\circ\det).$$

Main Theorem A: with the explicit construction of Klingen Eisenstein series in [Wan 15]

$$\frac{\left(\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi}\right)^2}{||\Phi||^2_{\sigma,\mathtt{MVW}}\,||\Psi||^2_{\pi,\mathtt{MVW}}} = \frac{1}{2^{\varkappa_\sigma+\varkappa_\pi}}\cdot \mathscr{L}_{\mathscr{V}^{\mathrm{ur}}_{\mathcal{F}}}(\sigma\times\pi) L_{\mathscr{U}^{\mathrm{ur}}_{\mathcal{F}}}\left(s+\frac{1}{2},\pi^\vee,\overline{\chi}^{\mathrm{c}}\right)^2$$
$$\times \prod_{v\in\mathscr{U}^{\mathrm{ur}}_{\mathcal{F}}} d_{N+1,v}(s,\overline{\chi}_v)^{-2} \prod_{v\in\mathscr{V}^{\mathrm{bad}}_{\mathcal{F}}} \mathscr{I}_v(\sigma_v,\pi_v) \prod_{v\in\mathscr{U}^{\mathrm{bad}}_{\mathcal{F}}} \mathscr{Z}_v^{\diamondsuit,\triangleright}(f^{\mathrm{Sieg}}_{v,s,\chi},\pi_v)^2,$$

Here the **local Ichino–Ikeda integral** is the local integral appearing in the Ichino–Ikeda formula:

$$\mathscr{I}_v(\sigma_v,\pi_v) = \frac{I(\Phi_v\boxtimes\Psi_v,\Phi_v^\vee\boxtimes\Psi_v^\vee)}{\langle\Phi_v\boxtimes\Psi_v,\Phi_v^\vee\boxtimes\Psi_v^\vee\rangle_{\Pi_v}},\quad \Phi_v\in\sigma_v,\ \Psi_v\in\pi_v,$$

$$I(\Xi_v,\Xi_v') := \int_{H(F_v)}\left\langle \Pi_v(\Delta^\sharp(h_v))\Xi_v,\Xi_v'\right\rangle_{\Pi_v}\mathrm{d}h_v,\quad \Xi_v\in\Pi_v:=\sigma_v\boxtimes\pi_v$$

The **local doubling-method integral** is the local integral defined and computed explicitly in [Wan 15].

II. GGP periods of Klingen Eisenstein series: Main Theorem A

In [Wan 15], the Klingen Eisenstein series is obtained by pulling back a Siegel Eisenstein series E_χ^{Sieg} on the larger group $G^\blacklozenge := \mathrm{U}(m+n+1, m+n+1)$ against $\Phi \in \sigma$. Using the embedding

$$\imath^\heartsuit : \mathrm{U}(m+1, n+1) \times (-\mathrm{U}(m,n)) \to \mathrm{U}(m+n+1, m+n+1),$$

one constructs

$$E^{\mathrm{Kling}}(F^\heartsuit(f_{s,\chi}^{\mathrm{Sieg}}, \Phi; -); g_0) := \int_{H(\mathcal{F})\backslash H(\mathbb{A})} E^{\mathrm{Sieg}}(f_{s,\chi}^{\mathrm{Sieg}}; \imath^\heartsuit(g_0, h))\overline{\chi}(\det h)\Phi(h)\, \mathrm{d}h$$

II. GGP periods of Klingen Eisenstein series: Main Theorem A

In [Wan 15], the Klingen Eisenstein series is obtained by pulling back a Siegel Eisenstein series E_χ^{Sieg} on the larger group $G^\blacklozenge := \mathrm{U}(m+n+1, m+n+1)$ against $\Phi \in \sigma$. Using the embedding

$$\imath^\heartsuit : \mathrm{U}(m+1, n+1) \times (-\mathrm{U}(m,n)) \to \mathrm{U}(m+n+1, m+n+1),$$

one constructs

$$E^{\mathrm{Kling}}(F^\heartsuit(f_{s,\chi}^{\mathrm{Sieg}}, \Phi; -); g_0) := \int_{H(\mathcal{F})\backslash H(\mathbb{A})} E^{\mathrm{Sieg}}(f_{s,\chi}^{\mathrm{Sieg}}; \imath^\heartsuit(g_0, h))\overline{\chi}(\det h)\Phi(h)\, \mathrm{d}h$$

1. First, substitute this construction into the GGP period integral and reduce it to a certain "cuspidal GGP period integral."

II. GGP periods of Klingen Eisenstein series: Main Theorem A

1. First, substitute this construction into the GGP period integral and reduce it to a certain "cuspidal GGP period integral."
2. Then square the integral and use the **Ichino–Ikeda formula** to factor it into a product of local integrals, involving local Ichino–Ikeda integrals and local doubling-method integrals.

II. GGP periods of Klingen Eisenstein series: Main Theorem A

1. First, substitute this construction into the GGP period integral and reduce it to a certain "cuspidal GGP period integral."
2. Then square the integral and use the **Ichino–Ikeda formula** to factor it into a product of local integrals, involving local Ichino–Ikeda integrals and local doubling-method integrals.
3. Finally, invoke the unramified computations of these local integrals. The local Rankin–Selberg and standard L-factors appear at the "good" places of $\mathcal{F}$, which include all but finitely many places.

II. GGP periods of Klingen Eisenstein series: Main Theorem B

We also computed the local integrals at the "bad places".

Main Theorem B (X., 2026): under technical assumptions, we have

$$\frac{\left(\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi}\right)^2}{||\Phi||^2_{\sigma,\mathtt{MVW}}||\Psi||^2_{\pi,\mathtt{MVW}}} = \frac{1}{2^{\varkappa_\sigma+\varkappa_\pi}}\cdot \mathscr{L}^{\mathscr{V}_{\mathcal{F}}^{(p)}}(\sigma\times\pi) L_{\mathscr{U}_{\mathcal{F}}^{\mathrm{ur}}}\left(s+\frac{1}{2},\pi^\vee,\overline{\chi}^{\mathsf{c}}\right)^2$$

$$\times \prod_{v\in\mathscr{V}_{\mathcal{F}}^{\mathrm{ram}}} \zeta_v C_{\sigma_v,\psi_v} C_{\sigma_v^\vee,\psi_v^{-1}} \mathcal{B}^{\mathrm{ess}}_{\pi_v}\mathcal{B}^{\mathrm{ess}}_{\sigma_v} \prod_{v\in\mathscr{V}_{\mathcal{F}}^{(p)}} \zeta_v\Delta_\beta^2\cdot \mathfrak{G}(\underline{\breve{\mu}},\underline{\breve{\lambda}})\mathfrak{G}(\underline{\breve{\mu}},\underline{\breve{\lambda}})\mathcal{C}^{\mathrm{ord}}_\sigma\mathcal{C}^{\mathrm{ord}}_\pi$$

$$\times \prod_{v\in\mathscr{U}_{\mathcal{F}}^{\mathrm{ur}}} d_{N+1,v}(s,\overline{\chi}_v^{\mathsf{c}})^{-2} \prod_{v\in\mathscr{U}_{\mathcal{F}}^{\mathrm{bad}}} \mathscr{Z}_v^{\diamondsuit}(f^{\mathrm{Sieg}}_{v,s,\chi},\pi_v)^2.$$

Here $\mathscr{L}^{\mathscr{V}_{\mathcal{F}}^{(p)}}(\sigma\times\pi)$ is the product of the corresponding local L-factors away from $\mathscr{V}_{\mathcal{F}}^{(p)}$, and the second line arises from the local Ichino–Ikeda integrals.

II. GGP periods of Klingen Eisenstein series: Main Theorems C and D

Let $\mathbf{f}$ (respectively $\mathbf{g}$) be an irreducible, tempered, cuspidal $\mathbb{I}$-adic (respectively $\mathbb{J}$-adic) Hida family on $\mathrm{U}(m,n)$ (respectively $\mathrm{U}(m+1,n)$).

II. GGP periods of Klingen Eisenstein series: Main Theorems C and D

Let $\mathbf{f}$ (respectively $\mathbf{g}$) be an irreducible, tempered, cuspidal $\mathbb{I}$-adic (respectively $\mathbb{J}$-adic) Hida family on $\mathrm{U}(m,n)$ (respectively $\mathrm{U}(m+1,n)$).

Main Theorem C (X., 2026)

Under certain technical assumptions including those above. Then there exists an element

$$\mathbf{P}_{\mathbf{G}} \in (\mathbb{I}^{\mathrm{ur}}[\![\Gamma_{\mathcal{K}}]\!] \otimes \mathrm{Frac}(\mathbb{I}^{\mathrm{ur}})) \otimes \mathrm{Frac}(\mathbb{J}^{\mathrm{ur}}),$$

such that for every "classical point" $\mathrm{P} \times \mathrm{Q} \in \mathcal{X}_{\mathbf{G}}^{\mathrm{gen}}$,

$$(\mathrm{P} \times \mathrm{Q})(\mathbf{P}_{\mathbf{G}}) = B(s_{\mathrm{P}}, \chi_{\mathrm{P}}) \mathcal{P}^{\mathrm{Kling}}(\mathbf{f}_{\mathrm{P}}, \mathbf{g}_{\mathrm{Q}}, \chi_{\mathrm{P}}, s_{\mathrm{P}}),$$

The right-hand side is the GGP-type period integral of a Klingen Eisenstein series against a cuspidal automorphic form, normalized by an explicit factor $B(s_{\mathrm{P}}, \chi_{\mathrm{P}})$ to make them algebraic.

$$\frac{\left(\mathcal{P}_{\Phi,\Psi}^{\mathrm{Kling}}\right)^2}{||\Phi||^2_{\sigma,\mathtt{MVW}}\,||\Psi||^2_{\pi,\mathtt{MVW}}} \approx \mathscr{L}^{\mathscr{V}_{\mathcal{F}}^{(p)}}(\sigma\times\pi)L_{\mathscr{U}_{\mathcal{F}}^{\mathrm{ur}}}\left(s+\frac{1}{2},\pi^\vee,\overline{\chi}^{\mathrm{c}}\right)^2$$

Main Theorem D (X., 2026)

Assume that the p-adic L-function $\mathcal{L}_p^{\mathscr{U}}(\mathbf{g},\chi_0)\in\mathbb{J}^{\mathrm{ur}}[\![\Gamma_{\mathcal{K}}]\!]\otimes\mathrm{Frac}(\mathbb{J}^{\mathrm{ur}})$ constructed in [Wan 15] is nonzero. Then there exists an element

$$\mathcal{L}_p(\mathbf{f}\times\mathbf{g})\in\mathrm{Frac}(\mathbb{I}^{\mathrm{ur}})\otimes\mathrm{Frac}(\mathbb{J}^{\mathrm{ur}})$$

such that for every $\mathsf{P}\times\mathsf{Q}\in\mathcal{X}_{\mathbf{G}}^{\mathrm{gen}}$,

$$(\mathsf{P}\times\mathsf{Q})(\mathcal{L}_p(\mathbf{f}\times\mathbf{g})) = \text{explicit local and normalization factors from Main Theorem B}$$
$$\times\,\mathscr{L}^{\mathscr{V}_{\mathcal{F}}^{(p)}}(\sigma_{\mathbf{f}_{\mathsf{P}}}\times\pi_{\mathbf{g}_{\mathsf{Q}}}).$$

III: Application to CM elliptic curves – Their Birch–Swinnerton–Dyer conjectures

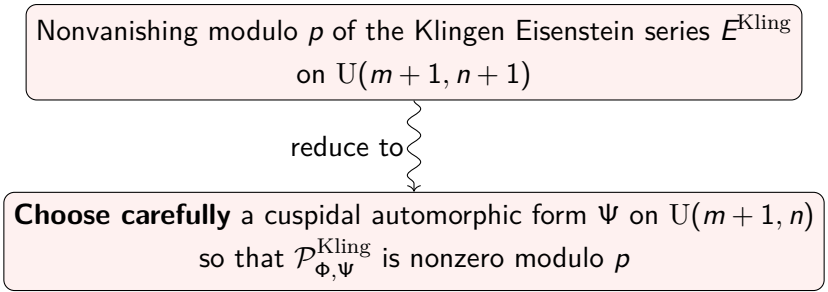
III. Application to the Iwasawa main conjecture for CM fields: general idea

By Main Theorem A:

Nonvanishing modulo p of the Klingen Eisenstein series E^{Kling} on $\mathrm{U}(m+1, n+1)$

III. Application to the Iwasawa main conjecture for CM fields: general idea

By Main Theorem A:



III. Application to the Iwasawa main conjecture for CM fields: general idea

By Main Theorem A:

Nonvanishing modulo p of the Klingen Eisenstein series E^{Kling} on $\mathrm{U}(m+1, n+1)$

reduce to

Choose carefully a cuspidal automorphic form Ψ on $\mathrm{U}(m+1, n)$ so that $\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi}$ is nonzero modulo p

reduce to

Choose carefully a cuspidal automorphic representation π on $\mathrm{U}(m+1, n)$ so that $L\left(\frac{1}{2}, \sigma \times \pi\right)$ and $L\left(s+\frac{1}{2}, \pi^{\vee}, \overline{\chi}^{\mathsf{c}}\right)^2$ are **simultaneously** nonzero modulo p

$H = \mathrm{U}(1,0)$: *Hecke characters* [Hsieh, 2014 JAMS] Here E^{Kling} is defined on $G^{\heartsuit} = \mathrm{U}(2,1)$.

- **Our GGP-period method**: Choose a cuspidal automorphic representation π of $G = \mathrm{U}(2,0)$:

$$\mathcal{P} = \frac{\left(\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi}\right)^2}{||\Phi||^2_{\sigma,\mathtt{MVW}}\,||\Psi||^2_{\pi,\mathtt{MVW}}} \approx \mathscr{L}^{\mathscr{V}^{(p)}_{\mathcal{F}}}(\sigma \times \pi) L_{\mathscr{W}^{\mathrm{ur}}_{\mathcal{F}}}\left(s + \frac{1}{2}, \pi^{\vee}, \overline{\chi}^{\mathsf{c}}\right)^2$$

This suggests nonvanishing modulo p results for these low-rank special L-values.

$H = \mathrm{U}(1,0)$: *Hecke characters* [Hsieh, 2014 JAMS] Here E^{Kling} is defined on $G^{\heartsuit} = \mathrm{U}(2,1)$.

- **Our GGP-period method**: Choose a cuspidal automorphic representation π of $G = \mathrm{U}(2,0)$:

$$\mathcal{P} = \frac{\left(\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi}\right)^2}{||\Phi||^2_{\sigma,\mathtt{MVW}}\,||\Psi||^2_{\pi,\mathtt{MVW}}} \approx \mathscr{L}^{\mathscr{V}^{(p)}_{\mathcal{F}}}(\sigma \times \pi) L_{\mathscr{W}^{\mathrm{ur}}_{\mathcal{F}}}\left(s + \frac{1}{2}, \pi^{\vee}, \overline{\chi}^{\mathsf{c}}\right)^2$$

This suggests nonvanishing modulo p results for these low-rank special L-values.

Our choice (taking σ to be trivial)

On $G = \mathrm{U}(2,0)$, choose π to be the **theta lift of a Hecke character** η from $G^{\natural} = \mathrm{U}(1,0)$ to G.

$H = \mathrm{U}(1,0)$: *Hecke characters* [Hsieh, 2014 JAMS] Here E^{Kling} is defined on $G^{\heartsuit} = \mathrm{U}(2,1)$.

- **Our GGP-period method**: Choose a cuspidal automorphic representation π of $G = \mathrm{U}(2,0)$:

$$\mathcal{P} = \frac{\left(\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi}\right)^2}{||\Phi||^2_{\sigma,\mathtt{MVW}}\,||\Psi||^2_{\pi,\mathtt{MVW}}} \approx \mathscr{L}^{\mathscr{V}^{(p)}_{\mathcal{F}}}(\sigma\times\pi) L_{\mathscr{W}^{\mathrm{ur}}_{\mathcal{F}}}\left(s+\frac{1}{2}, \pi^{\vee}, \overline{\chi}^{\mathrm{c}}\right)^2$$

This suggests nonvanishing modulo p results for these low-rank special L-values.

Our choice (taking σ to be trivial)

On $G = \mathrm{U}(2,0)$, choose π to be the **theta lift of a Hecke character** η from $G^{\natural} = \mathrm{U}(1,0)$ to G. Then

$$\mathcal{P} \approx \mathcal{P}(\eta,\lambda) := L^{\mathrm{alg}}\left(\frac{1}{2}, \lambda\eta\right) L^{\mathrm{alg}}\left(\frac{1}{2}, \lambda\right) L^{\mathrm{alg}}\left(s+\frac{1}{2}, \overline{\chi}^{-\mathrm{c}}\lambda^{-1}\eta^{-1}\right)^2 L^{\mathrm{alg}}\left(s+\frac{1}{2}, \overline{\chi}^{-\mathrm{c}}\lambda^{-1}\right)^2,$$

where λ is a Hecke character of $\mathcal{K}$ used as the "splitting character" in the theta-lift construction.

III. Application to the IMC for CM fields: The case $\mathrm{U}(1,0)$

Our choice (taking σ to be trivial)

On $G = \mathrm{U}(2,0)$, choose π to be the **theta lift of a Hecke character** η from $G^\flat = \mathrm{U}(1,0)$ to G. Then

$$\mathcal{P} \approx \mathcal{P}(\eta,\lambda) := L^{\mathrm{alg}}\left(\frac{1}{2}, \lambda\eta\right) L^{\mathrm{alg}}\left(\frac{1}{2}, \lambda\right) L^{\mathrm{alg}}\left(s+\frac{1}{2}, \overline{\chi}^{-\mathrm{c}}\lambda^{-1}\eta^{-1}\right)^2 L^{\mathrm{alg}}\left(s+\frac{1}{2}, \overline{\chi}^{-\mathrm{c}}\lambda^{-1}\right)^2,$$

where λ is a Hecke character of $\mathcal{K}$ used as the "splitting character" in the theta-lift construction.

Goal: We therefore seek λ and η such that $\mathcal{P}(\eta,\lambda)$ is nonzero modulo p.

III. Application to the IMC for CM fields: The case $\mathrm{U}(1,0)$

Our choice (taking σ to be trivial)

On $G = \mathrm{U}(2,0)$, choose π to be the **theta lift of a Hecke character** η from $G^\flat = \mathrm{U}(1,0)$ to G. Then

$$\mathcal{P} \approx \mathcal{P}(\eta,\lambda) := L^{\mathrm{alg}}\left(\frac{1}{2}, \lambda\eta\right) L^{\mathrm{alg}}\left(\frac{1}{2}, \lambda\right) L^{\mathrm{alg}}\left(s+\frac{1}{2}, \overline{\chi}^{-\mathrm{c}}\lambda^{-1}\eta^{-1}\right)^2 L^{\mathrm{alg}}\left(s+\frac{1}{2}, \overline{\chi}^{-\mathrm{c}}\lambda^{-1}\right)^2,$$

where λ is a Hecke character of $\mathcal{K}$ used as the "splitting character" in the theta-lift construction.

Goal: We therefore seek λ and η such that $\mathcal{P}(\eta,\lambda)$ is nonzero modulo p.

Tools: This is doable by a recent work of [Burungale–He–Tian–Ye] using ergodic theory in the theory of dymanic systems.

III. Application to the Iwasawa main conjecture for CM fields

Totally real field case: Recall in Part I, we introduced the Iwasawa main conjecture for class groups:

$$\mathrm{char}(\mathrm{Cl}_\infty) = (\mathcal{L}_p)$$

for the $\mathbb{Z}_p$-extension $\mathbb{Q}(\mu_{p^\infty}^+)/\mathbb{Q}(\mu_p)^+$ over the totally real field $\mathbb{Q}(\mu_p)^+$. This is a special case of the Iwasawa main conjecture for totally real number fields proved in [Mazur–Wiles, Wiles].

III. Application to the Iwasawa main conjecture for CM fields

Main Theorem E (X., 2026): the Iwasawa main conjecture for the CM field $\mathcal{K}/\mathcal{F}$

Assume $p \nmid 6D_{\mathcal{F}}|\Delta|$, and every place of $\mathcal{F}$ lying above p splits in $\mathcal{K}$. Then the following inclusion of Λ-ideals holds:

$$\mathrm{char}_{\Lambda}(X_{\Sigma}(\psi,\Sigma)) \subseteq (\mathcal{L}_{\mathtt{Katz}}(\psi,\Sigma)),$$

where $\psi : \Delta \to \overline{\mathbb{Z}}_p^{\times}$ is a branch character. [a finite-order Hecke character factoring through the finite group Δ].

Compared with [Hsieh 2014 JAMS], we remove most of its technical assumptions.

[Here $X_{\Sigma}(\psi,\Sigma)$ can be understand as the ψ-isotypic part of the "infinite class group" Cl_{∞} of the compositum of all $\mathbb{Z}_p$-extensions of $\mathcal{K}$.]

III. Application to the Iwasawa main conjecture for CM elliptic curves

Let E be an elliptic curve over $\mathcal{F}$ with complex multiplication by the ring of integers of an imaginary quadratic field $\mathcal{M}$.

- Assume that E has ordinary good reduction at every place of $\mathcal{F}$ above p. This implies that $p\mathcal{O}_{\mathcal{M}} = \mathfrak{p}\overline{\mathfrak{p}}$ splits in $\mathcal{M}$.
- Set $\mathcal{K} = \mathcal{F}\mathcal{M}$ and apply the Iwasawa main conjecture for the CM field $\mathcal{K}/\mathcal{F}$.

Main Theorem F (X., 2026): the Iwasawa main conjecture for CM elliptic curves over $\mathcal{F}$

Assume $p \nmid 6D_{\mathcal{F}}$. Then the following inclusion of ideals holds in $\Lambda_{E,\mathcal{O}_{\mathfrak{p}}}$:

$$\mathrm{char}_{\Lambda_E}(\mathrm{Sel}_{\mathfrak{p}^\infty}(E/\mathcal{K})^\vee) \subseteq (\mathcal{L}_p(E)).$$

III. Application to the Iwasawa main conjecture for CM elliptic curves

Let E be an elliptic curve over $\mathcal{F}$ with complex multiplication by the ring of integers of an imaginary quadratic field $\mathcal{M}$.

- Assume that E has ordinary good reduction at every place of $\mathcal{F}$ above p. This implies that $p\mathcal{O}_{\mathcal{M}} = \mathfrak{p}\overline{\mathfrak{p}}$ splits in $\mathcal{M}$.
- Set $\mathcal{K} = \mathcal{F}\mathcal{M}$ and apply the Iwasawa main conjecture for the CM field $\mathcal{K}/\mathcal{F}$.

Main Theorem F (X., 2026): the Iwasawa main conjecture for CM elliptic curves over $\mathcal{F}$

Assume $p \nmid 6D_{\mathcal{F}}$. Then the following inclusion of ideals holds in $\Lambda_{E,\mathcal{O}_{\mathfrak{p}}}$:

$$\mathrm{char}_{\Lambda_E}(\mathrm{Sel}_{\mathfrak{p}^\infty}(E/\mathcal{K})^\vee) \subseteq (\mathcal{L}_p(E)).$$

If, in addition, $p \nmid [\mathcal{F} : \mathbb{Q}]$ and $\mathcal{K}(E[\mathfrak{p}])/\mathcal{M}$ is abelian, then the equality holds.

III. Application to the BSD conjecture for CM elliptic curves

As a further application, we have the p-converse theorem and p-BSD formula for **rank-zero** CM elliptic curves over $\mathcal{F}$:

Main Theorem G (X., 2026)

Assume $p \nmid 6D_{\mathcal{F}}$ and that E has ordinary good reduction at every place above p. Then

$$\operatorname{corank}_{\mathbb{Z}_p} \mathrm{Sel}_{p^\infty}(E/\mathcal{F}) = 0 \Rightarrow L(E/\mathcal{F}, 1) \neq 0.$$

III. Application to the BSD conjecture for CM elliptic curves

As a further application, we have the p-converse theorem and p-BSD formula for **rank-zero** CM elliptic curves over $\mathcal{F}$:

Main Theorem G (X., 2026)

Assume $p \nmid 6D_{\mathcal{F}}$ and that E has ordinary good reduction at every place above p. Then

$$\mathrm{corank}_{\mathbb{Z}_p} \mathrm{Sel}_{p^\infty}(E/\mathcal{F}) = 0 \Rightarrow L(E/\mathcal{F}, 1) \neq 0.$$

If, in addition, $p \nmid [\mathcal{F} : \mathbb{Q}]$ and $\mathcal{K}(E[\mathfrak{p}])/\mathcal{M}$ is abelian, then $L(E/\mathcal{F}, 1) \neq 0$ implies $\mathrm{corank}_{\mathbb{Z}_p} \mathrm{Sel}_{p^\infty}(E/\mathcal{F}) = 0$ and

$$v_p\left(\frac{L(E/\mathcal{F}, 1)}{\Omega_E}\right) = v_p\left(\frac{|Ш(E/\mathcal{F})[p^\infty]|}{|E(\mathcal{F})_{\mathrm{tor}}|^2} \prod_{v \neq \infty} \mathrm{Tam}_v(E/\mathcal{F})\right).$$

IV: Future directions – My "mathematical life" in Vienna

IV. Future directions: applications to higher-rank unitary groups

A natural extension is to study Iwasawa theory for **rank-two unitary groups**:

- Pure signature unitary groups: $\mathrm{U}(1,1)$ and $\mathrm{U}(2,0)$,

IV. Future directions: applications to higher-rank unitary groups

A natural extension is to study Iwasawa theory for **rank-two unitary groups**:

- Pure signature unitary groups: $\mathrm{U}(1,1)$ and $\mathrm{U}(2,0)$,
- Mixed signature unitary groups: for rank-two unitary groups whose signatures vary among the archimedean places of $\mathcal{F}$.

IV. Future directions: applications to higher-rank unitary groups

A natural extension is to study Iwasawa theory for **rank-two unitary groups**:

- Pure signature unitary groups: $\mathrm{U}(1,1)$ and $\mathrm{U}(2,0)$,
- Mixed signature unitary groups: for rank-two unitary groups whose signatures vary among the archimedean places of $\mathcal{F}$.

Important instance: Let V be a two-dimensional skew-Hermitian space over $\mathcal{K}$ such that

- at one fixed archimedean place $\infty_0 : \mathcal{F} \hookrightarrow \mathbb{R}$, the skew-Hermitian space V_{∞_0} has signature $(2,0)$, and
- at every other archimedean place ∞_i $(i = 1, \ldots, d-1)$, the skew-Hermitian space V_{∞_i} has signature $(1,1)$.

IV. Future directions: applications to higher-rank unitary groups

Important instance: Let V be a two-dimensional skew-Hermitian space over $\mathcal{K}$ of signature $(2,0),(1,1),\ldots,(1,1)$ at archimedean places of $\mathcal{F}$.

Applications to elliptic curves over totally real fields $\mathcal{F}$: (Haidong Li, ongoing, 2025+)

Suppose $E/\mathcal{F}$ is an elliptic curve with good ordinary reduction at every place of $\mathcal{F}$ above p, and $\mathrm{ord}_{s=1} L(E/\mathcal{F},1) = 1$. Then the p-converse theorem and the refined p-part of the Birch–Swinnerton-Dyer formula for the elliptic curve $E/\mathcal{F}$ hold, provided that:

IV. Future directions: applications to higher-rank unitary groups

Important instance: Let V be a two-dimensional skew-Hermitian space over $\mathcal{K}$ of signature $(2,0),(1,1),\ldots,(1,1)$ at archimedean places of $\mathcal{F}$.

Applications to elliptic curves over totally real fields $\mathcal{F}$: (Haidong Li, ongoing, 2025+)

Suppose $E/\mathcal{F}$ is an elliptic curve with good ordinary reduction at every place of $\mathcal{F}$ above p, and $\mathrm{ord}_{s=1} L(E/\mathcal{F},1) = 1$. Then the p-converse theorem and the refined p-part of the Birch–Swinnerton-Dyer formula for the elliptic curve $E/\mathcal{F}$ hold, provided that:

- **the Iwasawa main conjecture for the mixed signature unitary group $\mathrm{U}(\mathrm{V})$ holds**,
- the corresponding results for analytic-rank-zero elliptic curves $E'/\mathcal{F}$ hold,
- .....

[See Xin Wan's ICCM 2018 report for motivations, and Li's article `arXiv: 2608.11969`.]

IV. Future directions: Explicit p-adic families on unitary groups

To pursue the higher rank case, with

$$\mathcal{P} = \frac{\left(\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi}\right)^2}{||\Phi||^2_{\sigma,\mathtt{MVW}}\,||\Psi||^2_{\pi,\mathtt{MVW}}} \approx \mathscr{L}^{\mathscr{V}^{(p)}_{\mathcal{F}}}(\sigma\times\pi)L_{\mathscr{U}^{\mathrm{ur}}_{\mathcal{F}}}\left(s+\frac{1}{2},\pi^{\vee},\overline{\chi}^{\mathsf{c}}\right)^2$$

the problem is reduced to choosing an appropriate Ψ over $\mathrm{U}(m+1,n)$ such that it is non-zero modulo-p.

IV. Future directions: Explicit p-adic families on unitary groups

To pursue the higher rank case, with

$$\mathcal{P} = \frac{\left(\mathcal{P}^{\mathrm{Kling}}_{\Phi,\Psi}\right)^2}{||\Phi||^2_{\sigma,\mathtt{MVW}}\,||\Psi||^2_{\pi,\mathtt{MVW}}} \approx \mathscr{L}^{\mathscr{V}^{(p)}_{\mathcal{F}}}(\sigma\times\pi)L_{\mathscr{U}^{\mathrm{ur}}_{\mathcal{F}}}\left(s+\frac{1}{2},\pi^{\vee},\overline{\chi}^{\mathsf{c}}\right)^2$$

the problem is reduced to choosing an appropriate Ψ over $\mathrm{U}(m+1,n)$ such that it is non-zero modulo-p.

"Philosophy" learnt from $\mathrm{U}(1,0)$-case: reduce the rank of unitary groups

Choose Ψ (over $\mathrm{U}(m+1,n)$) to be the **theta lifting** of an appropriately chosen $\Psi^{\sharp}$ over a lower rank unitary group $\mathrm{U}(m,n)$ (or even smaller unitary groups).

IV. Future directions: p-adic families of theta lifts on unitary groups

This requires **p-adic properties of theta liftings**, where many questions remain open:

- Suppose $\Psi^\sharp$ is "algebraic", when is a theta lifting $\theta(\Psi^\sharp)$ "algebraic", nonzero modulo p?

IV. Future directions: p-adic families of theta lifts on unitary groups

This requires **p-adic properties of theta liftings**, where many questions remain open:

- Suppose $\Psi^\sharp$ is "algebraic", when is a theta lifting $\theta(\Psi^\sharp)$ "algebraic", nonzero modulo p?
- Suppose $\Psi^\sharp$ is p-ordinary, when will the theta lifting $\theta(\Psi^\sharp)$ be p-ordinary (or can we control its p-adic slope)? Can we deform them into Hida families accordingly? .....

IV. Future directions: p-adic families of theta lifts on unitary groups

This requires ***p*-adic properties of theta liftings**, where many questions remain open:

- Suppose $\Psi^\sharp$ is "algebraic", when is a theta lifting $\theta(\Psi^\sharp)$ "algebraic", nonzero modulo p?
- Suppose $\Psi^\sharp$ is p-ordinary, when will the theta lifting $\theta(\Psi^\sharp)$ be p-ordinary (or can we control its p-adic slope)? Can we deform them into Hida families accordingly? .....

Existing works: (surely not an exhaustive list):

IV. Future directions: p-adic families of theta lifts on unitary groups

This requires **p-adic properties of theta liftings**, where many questions remain open:

- Suppose $\Psi^\sharp$ is "algebraic", when is a theta lifting $\theta(\Psi^\sharp)$ "algebraic", nonzero modulo p?
- Suppose $\Psi^\sharp$ is p-ordinary, when will the theta lifting $\theta(\Psi^\sharp)$ be p-ordinary (or can we control its p-adic slope)? Can we deform them into Hida families accordingly? .....

Existing works: (surely not an exhaustive list):

- $(\mathrm{O}(n), \mathrm{Sp}(2m))$: X. Zhang, 2022.

IV. Future directions: p-adic families of theta lifts on unitary groups

This requires ***p*-adic properties of theta liftings**, where many questions remain open:

- Suppose $\Psi^\sharp$ is "algebraic", when is a theta lifting $\theta(\Psi^\sharp)$ "algebraic", nonzero modulo p?
- Suppose $\Psi^\sharp$ is p-ordinary, when will the theta lifting $\theta(\Psi^\sharp)$ be p-ordinary (or can we control its p-adic slope)? Can we deform them into Hida families accordingly? .....

Existing works: (surely not an exhaustive list):

- $(\mathrm{O}(n), \mathrm{Sp}(2m))$: X. Zhang, 2022.
- $(\mathrm{U}(1,0), \mathrm{U}(n,0))$: Y. Lee, 2024. (We applied his work in the case $n = 2$ to obtain the main theorems above.)

IV. Future directions: p-adic families of theta lifts on unitary groups

This requires ***p*-adic properties of theta liftings**, where many questions remain open:

- Suppose $\Psi^\sharp$ is "algebraic", when is a theta lifting $\theta(\Psi^\sharp)$ "algebraic", nonzero modulo p?
- Suppose $\Psi^\sharp$ is p-ordinary, when will the theta lifting $\theta(\Psi^\sharp)$ be p-ordinary (or can we control its p-adic slope)? Can we deform them into Hida families accordingly? .....

Existing works: (surely not an exhaustive list):

- $(\mathrm{O}(n), \mathrm{Sp}(2m))$: X. Zhang, 2022.
- $(\mathrm{U}(1,0), \mathrm{U}(n,0))$: Y. Lee, 2024. (We applied his work in the case $n = 2$ to obtain the main theorems above.)
- $(\mathrm{U}(\mathrm{V}), \mathrm{U}(\mathrm{W}))$ for V of signature $(m, 0)$ and W of signature (n, n): Z. Liu, 2025.

IV. Future directions: p-adic families of theta lifts on unitary groups

This requires ***p*-adic properties of theta liftings**, where many questions remain open:

- Suppose $\Psi^\sharp$ is "algebraic", when is a theta lifting $\theta(\Psi^\sharp)$ "algebraic", nonzero modulo p?
- Suppose $\Psi^\sharp$ is p-ordinary, when will the theta lifting $\theta(\Psi^\sharp)$ be p-ordinary (or can we control its p-adic slope)? Can we deform them into Hida families accordingly? .....

Existing works: (surely not an exhaustive list):

- $(\mathrm{O}(n), \mathrm{Sp}(2m))$: X. Zhang, 2022.
- $(\mathrm{U}(1,0), \mathrm{U}(n,0))$: Y. Lee, 2024. (We applied his work in the case $n = 2$ to obtain the main theorems above.)
- $(\mathrm{U}(\mathrm{V}), \mathrm{U}(\mathrm{W}))$ for V of signature $(m, 0)$ and W of signature (n, n): Z. Liu, 2025.
- Our ongoing work with Zhibin Geng: generalize Lee's construction to $(\mathrm{U}(\mathrm{V}), \mathrm{U}(\mathrm{W}))$ where V has signature $(m, 0)$, both $\mathrm{V}^{\oplus r}$ and W lie in the same Witt tower $(m, r \geq 1)$.

IV. Future directions: p-adic families of theta lifts on unitary groups

This requires ***p*-adic properties of theta liftings**, where many questions remain open:

- Suppose $\Psi^\sharp$ is "algebraic", when is a theta lifting $\theta(\Psi^\sharp)$ "algebraic", nonzero modulo p?
- Suppose $\Psi^\sharp$ is p-ordinary, when will the theta lifting $\theta(\Psi^\sharp)$ be p-ordinary (or can we control its p-adic slope)? Can we deform them into Hida families accordingly? .....

Existing works: (surely not an exhaustive list):

- $(\mathrm{O}(n), \mathrm{Sp}(2m))$: X. Zhang, 2022.
- $(\mathrm{U}(1,0), \mathrm{U}(n,0))$: Y. Lee, 2024. (We applied his work in the case $n = 2$ to obtain the main theorems above.)
- $(\mathrm{U}(\mathrm{V}), \mathrm{U}(\mathrm{W}))$ for V of signature $(m, 0)$ and W of signature (n, n): Z. Liu, 2025.
- Our ongoing work with Zhibin Geng: generalize Lee's construction to $(\mathrm{U}(\mathrm{V}), \mathrm{U}(\mathrm{W}))$ where V has signature $(m, 0)$, both $\mathrm{V}^{\oplus r}$ and W lie in the same Witt tower $(m, r \geq 1)$.
- $(\mathrm{U}(1,1), \mathrm{U}(2,1))$: F. Iudica, 2024.

IV. Future directions: special L-values are nonvanishing modulo p

With

$$\mathcal{P} = \frac{\left(\mathcal{P}_{\Phi,\Psi}^{\mathrm{Kling}}\right)^2}{||\Phi||^2_{\sigma,\mathtt{MVW}}\,||\Psi||^2_{\pi,\mathtt{MVW}}} \approx \mathscr{L}^{\mathscr{V}_{\mathcal{F}}^{(p)}}(\sigma\times\pi)L_{\mathscr{V}_{\mathcal{F}}^{\mathrm{ur}}}\left(s+\frac{1}{2},\pi^{\vee},\overline{\chi}^{\mathsf{c}}\right)^2$$

Nonvanishing modulo p of special L-values in higher rank. Nonvanishing modulo p for special values attached to cuspidal representations and their Rankin–Selberg products is widely regarded as more subtle than the corresponding Iwasawa main conjecture. In genuinely higher rank, almost nothing is known.

Thank you!

The slide will be available soon on my webpage:

`https://ruichenx.github.io/`

Ruichen XU

`ruichen.xu@univie.ac.at`